

FDN 1202

Introduction to Computers

Computer Ethics

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Outline

- Ethics and Ethical Principles
- Ethics in Computer Usage
 - 10 Rules of Computer Ethics
- Concerns/Issues in Computer Ethics
 - Intellectual Property Rights
 - Patents
 - Copyrights
 - Software License
 - Software Piracy
 - Trademark
- Computer Crimes
 - Malware
 - Identity Theft
 - Cyber Stalking
 - Cyberterrorism
 - Spam



What are Ethics?

- Moral principles that govern a person's behavior or the conducting of an activity.
- Moral Principle – the principle of right and wrong that are accepted by an individual or a social group.



Ethical Principles

Ethical Principles are tools which are used to think through difficult situations

- The principle of Respect for autonomy
- The principle of Beneficence
- The Principle of non-maleficence
- The Principle of justice



The Principle of Respect for Autonomy

- Autonomy is Latin for “self-rule”
- We have an obligation to respect the autonomy of other persons, which is to respect the decisions made by other people concerning their own lives.
 - Corollary principles : Honesty in our dealings with others and obligation to keep promises.

The Principle of Beneficence

- We have an obligation to bring about good in all our actions
 - Corollary principles : We must take positive steps to prevent harm.

The Principle of non maleficence

- We have an obligation to not harm others
- Each action must produce more good than harm.
 - Corollary Principle:
 - Where harm cannot be avoided, we are obligated to minimize the harm we do.
 - Don't increase the risk of harm to others.
 - It is wrong to waste resources that could be used for good.

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- The Golden Rule : Do unto others as you would have them do unto you.
 - Meaning : One should treat others as one would like others to treat oneself



The Principle of Justice

- We have an obligation to provide others with whatever they are owed or deserve.
- In public life, we have an obligation to treat all people equally, fairly, and impartially.
 - Corollary principle : Impose no unfair burdens.

Ethics in Computer Usage

- Computer ethics deal with the procedures, values and practices that govern the process of consuming computing technology and its related disciplines without damaging or violating the moral values and beliefs of any individual, organization or entity.
- Simply computer ethics are set of moral principles that regulate the use of computers.

Ten Rules of Computer Ethics

1. You shall not use a computer to harm other people
2. You shall not interfere with other people's computer work.
3. You shall not snoop around in other people's computer files.
4. You shall not use a computer to steal.
5. You shall not use a computer to bear false witness.
6. You shall not copy or use proprietary software for which you have not paid (without permission).

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7. You shall not use other people's computer resources without authorization or proper compensation.
 8. You shall not appropriate other people's intellectual output.
 9. You shall think about the social consequences of the program you are writing or the system you are designing.
 10. You shall always use a computer in ways that ensure consideration and respect for your fellow humans.

Common issues of computer ethics

■ Privacy

- Responsibility to protect data about individual.

■ Accuracy

- Responsibility of data collectors to authenticate information and ensure its accuracy.

■ Property

- Who owns information and software and how can they be sold and exchanged.

■ Access

- Responsibility of data collectors to control access and determine what information a person has the right to obtain about others and how the information can be used.

■ Example Ethic Violations:

- While it is easy to duplicate (copyrighted electronic or digital) content, computer ethics would suggest that it is wrong to do so without the author's approval.
- While it may be possible to access someone's personal information on a computer system, computer ethics would advise that such an action is unethical.

Intellectual Property Rights

- Intellectual property refers to creations of the mind, such as inventions, literary and artistic works, designs, symbols, names and images used in commerce.
- Intellectual property rights are the protections granted to the creators of IP, and include
 - Trademarks
 - Copyright
 - Patents
 - Industrial design rights
 - Trade secrets



Patent

- A patent is a form of right granted by the government to an inventor, giving the owner the right to exclude others from making, using, selling, offering to sell, and importing an invention for a limited period of time, in exchange for the public disclosure of the invention.
- That product or a process should provide, in general an new way of doing something, or offers a new technical solution to a problem.



Copyright

- Copyright (or author's right) is a legal term used to describe the rights that creators have over their literary and artistic works.
- A copyright gives the creator of an original work exclusive rights to be the only one to reproduce, publish, and sell, usually for a limited time.
- Works covered by copyright range from books, music, paintings, sculpture, and films, to computer programs, database advertisements, maps, and technical drawings.

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- Software developers (or the companies they work for) own their programs.
 - Software buyers only own the right to use the software according to the license agreement.
 - No copying, reselling, lending, renting, leasing, or distributing is legal without the software owner's permission.

Software License

- A software license is a legal instrument allowing consumers to use or redistribute software.
- Using the software without the license agreement, would constitute a breach of copyright law.
- The license agreement explains to the end-user how they can use the software
- There are four types of software licenses:
 - Public Domain
 - Freeware
 - Shareware
 - All Rights Reserved

Software License

■ Public Domain

- Public domain software has no owner and is not protected by copyright law.
- It was either created with public funds.
- Can be copied, sold, and/or modified.

■ Freeware License

- Freeware is copyrighted software that is licensed to be copied and distributed without charge.
- Freeware is free, but it's still under the owner's control.

■ Shareware License

- A shareware software license allows you to use the software for a trial period, but you must pay a registration fee to the owner for the permanent use.

■ All Rights Reserved License

- May be used by the purchaser according the exact details spelled out in the license agreement.
- You can't legally use it--or even possess it-- without the owner's permission.

Privacy vs Piracy

■ Privacy

- The degree to which an individual can determine which personal information is to be shared with whom and for what purpose.
- A violation of privacy exposes your personal information to unauthorized ones.

■ Piracy

- The unauthorized use or reproduction of another's work.

■ Software Piracy

- Produces exactly the same or similar version without giving proper credit to the original author or simply produces it and distributes it to others.

Types of software piracy

■ Softlifting

- Borrowing and installing a copy of a software application from a colleague.

■ Client-server overuse

- Installing more copies of the software than you have licenses for.

■ Hard-disk loading

- Installing and selling unauthorized copies of software on new computers.

■ Counterfeiting

- Duplicating and selling copyrighted programs.

■ Online piracy

- Typically involves downloading illegal software from peer-to-peer network, Internet action or blog.

Trademark

- A trademark is a sign, design or expression capable of distinguishing the goods or services of one enterprise from those of other enterprises.
- Trademarks date back to ancient times when craftsmen used to put their signature or "mark" on their products.



Computer Crime

- Alternatively referred to as cyber crime, e- crime, electronic crime, or hi-tech crime.
- Computer crime is an act performed by a knowledgeable computer user, sometimes referred to as a hacker that illegally browses or steals a company's or individual's private information.
- In some cases, this person or group of individuals may be malicious and destroy or otherwise corrupt the computer or data files.

■ E.g.

- Malware.
- Identity Theft.
- Cyberbully or Cyberstalking.
- Cyber terrorism.
- Spamming.
- Denial of Service attack.
- Phishing.

Malware

- Alternatively referred to as scumware, malware is malicious software designed to change your settings, delete software, cause errors, watch browsing habits, or open computer to attacks.
- Different types of malware
 - Virus
 - Worms
 - Trojan Horses
 - Rootkits
 - Spyware
 - Keylogger



Different types of Malware



■ Virus

- A computer malware that can spread across computers and network by making copies of itself without the user's knowledge.
- A virus program has to be run before it can infect a computer.
- Technically a virus is a program or code that attaches itself to a legitimate, executable piece of software.
- The first virus known as the Elk Cloner was written by Rich Skrenta in 1982.
- E.g.
 - Appending virus
 - CMOS virus
 - Macro virus
 - Boot sector virus
 - Encrypted virus



■ Worms

- Similar to viruses but do not need a carrier program or document.
- Worms simply create exact copies of themselves and use a network to spread.
- It uses a computer network to spread itself, and it relies on network loophole and security failure to travel from one host to another automatically and usually don't requires user intervention

E.g.

- Morris
- Code Red
- MSBlast
- Blaster



■ Trojan horses

- Programs that pretend to be legitimate software, but actually carry out hidden, harmful functions.
- Trojans cannot spread as fast as viruses because they do not make copies of themselves.
- E.g.
 - Zeus trojan



■ Spyware

- Spyware is another type of malware which collects data from your PC without your knowledge and permission.
- Spyware runs in the background and collects your personal information like your browsing pattern, sites you visit, email, cookies, saved data into browsers, website passwords and even credit card details.

■ Rootkits

- A rootkit is a collection of tools that enable administrator level access to a computer or computer network.

■ Keylogger

- Monitors your keystrokes, logging them to a file and sending them off to remote attackers.

Identity Theft

- Pretending to be someone you are not.
- Someone gains access to your personal information and uses it for their own benefit.
 - Stealing your online banking account login and password to getting access and using it to pretend to be you.
 - Such people can make themselves a lot of money with your personal information.

Cyberstalking

- Harassing others online.
- People leave a lot of information about themselves online. Such information can leave you vulnerable to cyberstalking.

Cyber Terrorism

- Hacking, threats and blackmailing towards a business or a person

Spam

- Distributed unsolicited e-mail to dozens or hundreds of different addresses.

Home work

What are the different strategies that can be used to protect yourself from computer crimes?

Thank You
